Project ref.: **N5-20261015-095736-0B9021AD** | Issued: **2026-10-25** | **International Edition (INT)**

Normative compliance: **Codes and standards applicable at the project site, to be stated by the responsible engineer**

⚠ **Investment, maintenance and electricity costs were not informed: the economic result shows only the savings on the water bill; payback and NPV are not meaningful until costs are entered.**

⚠ **The grid emission factor was not informed: the emissions from pumping electricity are not meaningful until it is entered.**

EXECUTIVE TECHNICAL REPORT (A4 LANDSCAPE)

Triton — Rainwater Harvesting | Project: **Example - tutorial house in Cape Town (ERA5 1991-2020) [costs to be entered by the user]**

CLIMATE SITE	CLIMATE DATASET	SERIES PERIOD	ECONOMIC HORIZON
Cape Town / South Africa	Observed series	01/01/1991 – 31/12/2020	20 years

RECOMMENDED VOLUME	POTABLE-WATER SAVINGS	NPV (20 YEARS)	ESTIMATED PAYBACK
20,000 L	**19.4%**	**Not calculated**	**Not calculated**
Lower Cistern	Mains-Water Replacement	Net Present Return	Not calculated (discounted)

1. BUILDING & HYDROLOGY

Technical Parameter	Value
Rainfall Series	**CapeTown.csv**
Catchment Area	**150.0 m²**
Runoff Coeff. (C)	0.85
First-Flush Discard	2.0 mm
Population Served	4 people
Target Rainwater Demand (28%)	**168 L/day**

2. SIMULATED WATER BALANCE

Operational Indicator	Performance
Cistern Volume	**20,000 L**
Potable-Water Savings	**19.4%**
Mean Rainwater Use	**116.5 L/day**
Mean Potable-Water Use	483.5 L/day
Overflow Volume	5.1 L/day
Full Supply (100%)	**68.2% of days**
No Supply (0%)	**29.6% of days**

3. ECONOMIC FEASIBILITY

Financial Indicator	Result
CAPEX Investment	**Not calculated**
Gross Savings, Year 1	**R 3,579.88 / yr**
Simple Payback	Not calculated
Discounted Payback	Not calculated
NPV (20 Years)	**Not calculated**
IRR (% p.a.)	**Not calculated**

📌 Technical Opinion on Water Engineering & Sustainability

⚠ Technical Responsibility & Validation Notice

Triton is a computational sizing-support tool. **The licensed professional is responsible for** validating the executive assumptions, network interferences and compliance with local legislation.

Under the recorded assumptions, the rainwater system shows modelled potable-water savings of **19.4%**; the economic feasibility was not calculated in this study. Sizing follows the technical standards and regulations applicable at the project site.

Rainfall: weather data from Open-Meteo.com (CC BY 4.0), based on the ERA5 reanalysis of the Copernicus Climate Change Service (Hersbach et al., 2020).

Engineer in Charge / Professional
Professional registration / Licence number